

Limiting the Threats from Autonomous Nuclear Weapons Launch

Group 5: Zoe Gastelum, Hebah Malik, Robert Mbenoun Mahend, Emmanuel Mate-Kole, Shang Wen

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1.0 Introduction, Background, and Scope

The nuclear nonproliferation regime has done a remarkable job at limiting both the horizontal spread of nuclear weapons and their related systems across the globe. The absence of use of nuclear weapons in military conflict since 1945, the somewhat muted numbers of nuclear weapons states, and spread of peaceful nuclear technologies and applications are a testament to the regime's effectiveness. However, we are in an age of rapid technological development and expansion, and the nonproliferation regime needs to be updated to account for emerging technologies and the opportunities and threats they may pose for nuclear nonproliferation. One of the most rapidly expanding areas – and thus the topic of our paper – is autonomy. Bolstered by advances in computational hardware and software, autonomous technologies that once seemed like science fiction are becoming a reality. While no state has (to our knowledge) yet incorporated autonomous functionality into the nuclear weapons command and control systems, the time is ripe to mutually agree the actions needed to be taken on fully autonomous launch capabilities for nuclear weapons to increase trust, limit potential for miscommunication and accident, and demonstrate good faith towards our nonproliferation agreements.

There is a gap in the current nonproliferation and arms control regime, in which there are no agreed-upon limits on the development and integration of autonomy technologies into nuclear weapons launch capabilities. The objective of this research is to evaluate policy options for the United States Government to pursue leadership in limiting autonomous nuclear weapons launch both domestically and internationally. Using multi-attribute utility analysis, we find that a multilateral, non-binding agreement is most feasible given the current state of geopolitics and technology, with the hope that such an agreement will bring key players to the negotiating table as part of a longer-term effort towards a binding international treaty.

1.1 Theory

Our understanding of the risks of adopting autonomous nuclear weapons launch system is largely informed by concept of security dilemma, originated from the realist paradigm of international relations theories. The security dilemma refers to one state's action to increase its own security – in this case, the decision to implement autonomy the not only in the rest of the command-and-control system of its nuclear weapon but specifically autonomous launch – could cause reactions from other states, which is thought to inevitably decrease the security of the first mover, causing an arms race spiral.

Within this paradigm, the race towards the first actor with autonomous launch capability is risky. From the perspective of deterrence theory, incorporating launch autonomy could make nuclear

deterrence more credible, as the would-be nuclear attack is guaranteed to be met with retaliation in a very short period of time, provided one could survive first strike, which has long been the argument of the proponents of the hair-trigger warning or launch-on-warning/detection policy.

However, from the same theoretical perspective, autonomous launch is problematic in several ways. First, it undermines the concept of mutual deterrence, which is a key component of the nuclear arms race. Mutual deterrence relies on the assumption that both sides possess the capability to retaliate in the event of a nuclear attack, which prevents either side from launching a first strike. Autonomous launch (especially mutually-assured autonomous launch) eliminates this assumption and makes it impossible for either side to avoid catastrophic consequences, regardless of who strikes first. Secondly, the autonomous launch creates a situation where rational decision-making process breaks down. In game theory, the most rational decision is often not to act, but the autonomous launch removes this option by ensuring that any attack will result in total destruction. This creates a situation where neither side has any incentive to refrain from launching an attack, which could result in an accidental nuclear war. Furthermore, the implementation of an autonomous nuclear launch system could increase the perceived risks of a first strike. If a country believes that its adversary has an automated system in place, it may feel compelled to launch a preemptive strike to prevent the system from being activated. This could lead to an arms race and a dangerous escalation of tensions between countries.

Therefore, before any major nuclear states successfully install an autonomous nuclear launch system, it is imperative for the international community, especially the U.S., Russia, and China to develop mechanisms to preemptively avert technological and policy escalations.

2.0 Literature Review

2.1 Command & Control and Human Factors

Nuclear command and control (NC2) are the processes and procedures that are in place for the U.S. president to launch a nuclear strike. In the United States, the President has sole authority to decide to use nuclear weapons. There is some discussion in academia and think tanks about changing the sole authority – for example, requiring the President to consult with the Vice President, Speaker, or legal council before launching a first use attack when there is more ample time than in a response attack. This posture of sole authority has historically been justified by the need to be able to respond quickly to the warning of an incoming nuclear attack.

While the US President still retains sole authority, there are variations on how that authority may be carried out. For instance, one strategy is known as “launch on attack” also known as “launch on warning.” The launch on warning strategy was adopted in the 1970s, and gives approval for a nuclear response as soon as an incoming nuclear strike is detected/confirmed. The detection mechanisms may rely on computerized systems, and is vulnerable to misidentification of an incoming attack such as the 1983 false alarm received by the Soviet military that has made Stanislav Petrov known as “the man who saved the world” (Chan, 2017). Once the president gives authorization for a nuclear attack, it can take approximately 5-10 minutes before missiles are launched. Once a nuclear attack is launched, it cannot be recalled (Moniz & Nunn, 2019). Launch on warning could give the misperception that weapons are poised for first use, which

could be a dangerous geopolitical signal itself. Russia is believed to have a hair trigger alert and as of 2021 China was believed to be preparing such a capability (von Hippel, 2021). Both President H.W. Bush and Obama both considered removing the launch on warning policy, but did not do so.

In addition to geopolitical messaging around automation in command and control, hair trigger responses also should be considered within the context of human factors. Human factors is a field that accounts for human capabilities and limitations in the design of products and systems. Most relevant for nuclear command and control, human factors tries to understand why humans fail or make errors in some situations, and then design systems to avoid those potential errors. Automation is seen as a way to limit human errors in some instances, for which there are multiple examples. But human errors are not the only errors with which we should be concerned related to nuclear command and control. Automation can sometimes fail in unexpected ways. For example, in a 2018 Uber's self-driving vehicle software resulted in a crash that killed a pedestrian walking a bicycle because it was unable to identify if the pedestrian was a vehicle, bicycle, or pedestrian. The classification issue inhibited its ability to predict their path, delayed braking, and ultimately resulted in the pedestrian's death (Marshall & Davies, 2019)

2.2 Autonomy and Military Weapons Deployment Launch Current Status

An autonomous weapon consists of a system that can select and engage targets without human intervention. In the U.S. policy directive on autonomous weapons (U.S. Department of Defense (DOD), 3000.09, updated 01/25/2023), the DOD describes the responsibilities and guidelines associated with the development and deployment of semi-autonomous to fully-autonomous weapons.¹ While the roster of fully autonomous weapons systems in the United States is limited, there are current naval defense weapons such as the U.S. Phalanx gun system that can fire autonomously when a ship is under attack by enemy missiles. However, current systems are used in defense only and do not independently search for or strike enemy assets.

In addition to its development of guidelines for autonomous weapons development and deployment, the DOD announced in February 2020 the establishment of five responsible artificial intelligence (AI) principles that will serve as scaffolding to autonomous weapons systems, including: responsible, equitable, traceable, reliable, and governable.² DOD has published other directives, memos, and statements related to its use of autonomous conventional weapons indicating their acceptable use for defensive purposes.³ However, the DOD directive on autonomous weapons systems does not specify policy surrounding autonomy and nuclear weapons.

2.3 Existing Nonproliferation Frameworks and Gaps

The existing international frameworks in the context of the policy options presented are the Measures for the Further Reduction and Limitation of Strategic Offensive Arms (New START Treaty), the United Nations Security Council (UNSC) 1540 Committee, the Treaty on the Prohibition of Nuclear Weapons (TPNW), and the Nuclear Nonproliferation Treaty (NPT).

2.3.1 New START

The New START Treaty was the last treaty between the Russian Federation and the United States that provided transparency and limitations on both parties' nuclear arsenal. The New

START treaty was suspended in February 2023, and was the last standing effort to provide lines of effective and transparent communication between Russia and the United States. While it was a bilateral agreement, there were a few gaps to the New START Treaty such as effective verification as well as limiting the testing of new nuclear weapons and delivery systems. With this treaty, the United States and Russia were able to continue development of fully autonomous deployment weapons without further discussion between the two parties. The New START Treaty's gap was unable to update with the evolving emerging technology and could not have provided effective communication regarding emerging technology such as fully autonomous nuclear weapons deployment.

2.3.2 The UNSC 1540 Committee

2.3.3 Treaty of the Non-Proliferation of Nuclear Weapons

The Treaty on the Non-Proliferation of Nuclear Weapons (or Nonproliferation Treaty, NPT), opened for signature in 1968 and entered into force in 1970 (United Nations Office for Disarmament Affairs, n.d.). The main tenants of the treaty are that 1) nuclear weapon states not transfer nuclear weapons technology to non-nuclear weapon states; 2) non-nuclear weapon states agree not receive or otherwise acquire nuclear weapons; and 3) that nuclear weapon states pursue good faith negotiations for nuclear disarmament (International Atomic Energy Agency, 1970). The NPT also requires that non-nuclear weapon states sign nuclear safeguards agreements with the International Atomic Energy Agency (IAEA) to verify of its peaceful use of nuclear materials and facilities. While the NPT does not explicitly require it, the nuclear weapon state signatories of the NPT have also accepted safeguards agreements for selected civilian aspects of their nuclear programs as a measure to not give their domestic nuclear energy industries advantage over non-nuclear weapon states. There are currently 191 state parties to the NPT (United Nations Office for Disarmament Affairs, n.d.) India, Pakistan, India and South Sudan have not signed the NPT, and the Democratic People's Republic of Korea (DPRK, i.e., North Korea) withdrew from the treaty in 2003 (Center for Arms Control and Nuclear-Nonproliferation).

The NPT is often considered foundation of the international nuclear nonproliferation regime. Through their signature of the NPT and subsequent safeguards agreements with the IAEA, non-nuclear weapon states have provided transparency and confidence that they are not developing nuclear weapons programs for their neighbors, allies, and adversaries. Following the revelation of undeclared nuclear programs in Iraq and DPRK in the early 1990s, the IAEA published a Model Additional Protocol (i.e., Additional Protocol, AP) for states to provide even more access, reporting, and transparency on their nuclear programs to provide the IAEA sufficient legal basis to verify the absence of undeclared nuclear facilities in a state (International Atomic Energy Agency, 2022).

The NPT represents a broad-based, multilateral treaty signed by sovereign states, and includes significant participation by both nuclear weapon states and non-nuclear weapon states. The inclusion of IAEA safeguards as a treaty-mandated verification mechanism allows violations of the treaty to be referred to the United Nations Security Council and the General Assembly (International Atomic Energy Agency, 1989). The NPT is not intended to cover arms control or nuclear weapon dismantlement activities, for several reasons. First, it was intended for the

nuclear weapon state signatories of the NPT to come to their own arrangements based on the negotiations described in the NPT's Article VI. Despite the discontent of some states in the international community, the United States and Russia (and before that the USSR) have been working bilaterally to reduce nuclear weapon delivery systems and deployed nuclear warheads via arms control agreements such as the New START agreement (currently suspended by Russia). Second, verification of weapons-related information could risk violating Article I of the NPT, in which nuclear weapon states agree not to transmit such information to non-nuclear weapon states, which includes member states of the IAEA. In addition to typical avoidance of weapons-related information, current verification regimes for the NPT under IAEA safeguards and Additional Protocols does not examine verification of software, such as autonomous nuclear weapon launch software, that could be part of a future agreement to limit autonomy and nuclear weapons.

2.3.4 Treaty on the Prohibition of Nuclear Weapons

Under the Treaty on the Prohibition of Nuclear Weapons, signatories agree not to possess or acquire nuclear weapons or other nuclear explosive devices, to transfer or receive such devices, to use or threaten use of them, to seek them, or to have nuclear weapons deployed or stationed in their territory (United Nations, 2017). The treaty requires that states with nuclear weapons irreversibly eliminate their nuclear weapons programs, which is to be verified by the “competent international authority” (United Nations, 2017), and that all states maintain safeguards agreements with the International Atomic Energy Agency for ongoing monitoring of their compliance with this agreement, via comprehensive safeguards agreements that have already been established through INFCIRC/153 (the safeguards agreements established under the Nuclear Nonproliferation Treaty).

The TPNW was unique in the role that nongovernmental organizations and members of civil society played in its adoption. The International Campaign to Abolish Nuclear Weapons (ICAN) played a significant role in the treaty's adoption, which took place in July 2017 (ICAN, n.d.). As of 2023, there are 92 signatory states, 68 of which have ratified the treaty. However, no Nuclear Weapons State has participated in the negotiation of the treaty nor has signed.

3.0 Policy Options

Based on our assessment of the current nuclear nonproliferation regime and understanding of autonomy technologies, we developed six policy options:

1. Status quo. Under the status quo, the United States will take no action to limit the development of autonomy for our nuclear weapons program. Leaving open the option to develop fully autonomous nuclear weapons launch will allow us to attempt to gain parity with China, seen as the current leader in this area. Permitting free growth in this area could support the emergence of US leadership in autonomy technologies, which has security benefits. This option could lead to an arms race with China, Russia, or others, which could be both costly and dangerous. The quick pace of technological development could pose unforeseen cyber risks or jump ahead of the regulation curve.
2. Unilateral action. The United States could take unilateral action to limit our development of autonomous nuclear launch. We could frame this as an international security issue on

which we are taking an individual stand and invite other countries that we see as technologically capable in this area to make similar statements. Unilateral limits are consistent with the Department of Defense's current posture in which only humans make decisions about deploying lethal force. Further, unilateral limits could reduce cyber risks related to integration of overly complex systems into our nuclear weapons launch protocols. Limited autonomy research for nuclear launch could save research funding to be reinvested in more pressing areas such as climate change. However, unilateral action could allow the United States to be technologically overcome by autonomy research in adversary countries.

3. Bilateral agreement with Russia. Bilateral agreements could potentially be negotiated with Russia in a similar style to previous bilateral nuclear arms accords. Despite our history of arms control agreement with Russia, the war in Ukraine and suspension of the New START Treaty make engagement with Russia significantly more difficult than usual. In addition, a bilateral agreement typically could include some type of verification measure, which would pose new technical challenges if computer code is to be verified.
4. Bilateral agreement with China. Bilateral agreements with China could follow a similar model to those with Russia. While China has historically not been interested in participating in arms control agreements with the United States and Russia due to its smaller nuclear arsenal, its advanced capabilities with respect to autonomy might present such an opportunity.
5. Multilateral agreement. A multilateral agreement against the development of fully autonomous nuclear launch could represent an international dialogue on the issue. A multilateral agreement would not likely be legally binding, and could potentially have light or no verification measures, for example based on the annual self-reporting that states submit under their United Nations Security Council Resolution 1540 obligations, or groups of experts for conventional arms control measured under the United Nations. Though a multilateral agreement could proceed without Russia or China, it is not clear if it would have impact without these key players.
6. Binding treaty. A binding treaty could make illegal the use of full autonomy for nuclear weapons launch. The treaty would be accompanied by formal verification measures carried out by a group of members or a designated third party. A binding treaty would have international legal enforcement, for which a state not complying with the treaty could be sanctioned by the United Nations. The verification measures could build confidence among treaty partners that autonomous launch is not needed. While it is possible to have treaties like this enter into force, unless the United States, Russia, and China all sign and ratify (which in the present is incredibly unlikely due to geopolitical tensions) the treaty will be missing the major players and have limited impact. Furthermore, opening the US command and control system to international inspections would require much more cyber risk than bilateral inspection regimes with a single nuclear-armed partner.

4.0 Methodology:

To assess these options, we selected the multi-attribute utility analysis (MAUA) approach. MAUA defines a utility function (see Equation 1) in which each option is assessed based on various attributes. Each attribute score is weighted according to its respective importance to overall utility. The weighted attribute scores are summed to produce an overall utility score.

Equation 1:

$$\text{Utility}_0 = \sum_{i=1}^n \text{weight}_i \cdot \text{attribute}_{i,0}$$

We defined four relevant attributes by which to judge each policy option. The attributes include geopolitical risks and difficulty, norms and treaties, technical verification measures, and cyber concerns. Each of these attributes is described in additional detail below. We used consensus-based decision-making to weight the attributes and to score each policy option. Consensus-based means that we discussed the weights and scoring as a group. While we could have individually scored the regime and calculated the mean of the weights or scores, we thought that consensus-based scoring was less likely to inadvertently mask a disagreement among team members.

We weighted the attributes as follows:

- Geopolitical: 55%
- Norms: 20%
- Verification: 15%
- Cyber: 10%

Each policy option was assessed on the attributes using a Likert scale from one to five, where 5 represents the hardest or riskiest option, and 1 represents the least risky or easiest option.

4.1 Cyber Risks

For autonomous nuclear weapon systems, AI along with sensors and other technologies are needed for sophisticated capabilities, such as obstacle detection and maneuverability, automated target identification, and longer-range and loitering capability. Today's technology and algorithms face challenges in reliably naming objects, responding in real time, planning, and controlling routes in the absence of GPS, and defending against cyberattacks. Given the lack of technological maturity, fully autonomous nuclear-weapon systems are highly risky. These risks, combined with the potential instability these weapons may cause, suggest that a ban on fully autonomous systems is called for until the technology is better understood and proven. With all the advancements in autonomy and machine learning, a broader range of physical systems are more vulnerable to cyber-attacks, including hacking, spoofing and data poisoning. Though these are great innovations, they have added a new and potentially more sinister twist to the risk of miscalculation, misperception, and inadvertent escalation that originates in cyberspace but has a very real world in the physical world.

4.2 Geopolitical Risks

The two countries we will be examining in the scope of geopolitical risks are China and Russia due to their increased nuclear capabilities and focus on technological expansion of fully

autonomous weapon launch. In regard to Russia, President Putin has emphasized that “digital technologies and AI, robotization, and unmanned systems should be on the qualitative development agenda of our Armed Forces”. Given their vehement position against Lethal Autonomous Weapons Systems (LAWS) in the CCW, it sets to develop as much as they can in the AI field.⁴ In addition, China’s further advancement has set goals to become the forefront of AI innovation and advancement. China’s largest military weapon manufacturer has set out that by 2025, lethal autonomous weapons would be commonplace by 2025.⁵ With their set goals of modernizing their nuclear arsenal, while it is not transparently clear how advanced they intend their autonomous research and development to be on lethal weapons, there is a possibility that the expansion of their nuclear arsenal and their intentions to be the forefront of AI innovation could result in the creation of a fully autonomous launch mechanism for their nuclear weapons. Given the geopolitical location of Russia and China, there is the possibility that their creation of a fully autonomous launch of nuclear weapons will cause other nuclear weapon states to establish the similar technology. This poses a threat in escalation and hostility amongst nuclear weapon states and can result in the deployment of nuclear weapons that omit the presence of a human. As AI enters the nuclear nonproliferation regime, evaluating the cyber and geopolitical risks is vital in order to draft a treaty regarding the banning of fully autonomous nuclear weapons launch.

4.3 Norms

The world has come a long way in efforts with nuclear arms control. Other than the multilateral treaty networks under the UN framework mentioned in the previous sections, the U.S. also had a rich history in bilateral engagement on the subject, especially with the Soviet Union/Russia. Following the Cuban Missile Crisis, the U.S. and the Soviet Union entered the era of détente and ushered in a more hopeful era of nuclear arms control, with the Limited Test Ban Treaty, Strategic Arms Limitation Talk (SALT), the Anti-Ballistic Missile (ABM) Treaty, and SALT II as the main platforms to slow down the pace of nuclear arms race. (Council on Foreign Relations) Furthermore, the Reagan and Bush administrations in the 80s and 90s engaged extensively with their Moscow counterparts on nuclear weapons and delivery vehicles reduction. However, as opposed to the multilateral agreements, this bilateral approach is more sensitive to the stability of the geopolitical situations between the two countries involved. As the Russo-American relationship deteriorated over the years, the Trump administration withdrew the U.S. from both the Intermediate-Range Nuclear Forces Treaty and the Treaty on Open Skies, as Washington repeatedly accused Russia of abusing these agreements. Similarly, the New START treaty is now in jeopardy, as aforementioned.

On the other hand, bilateral engagement with China appears to be much more difficult. As a prominent nuclear state and a permanent member of the UN Security Council, China only displayed interests in multilateral dialogues on the subject. Even so, China’s recent nuclear modernization and buildup efforts have been subject to intense controversy and accusations of violation of the NPT, which it had signed and ratified. Moreover, Beijing has yet ratified the CTBT and is not a signatory of the TPNW. (Nuclear Threat Initiative, 2022) While China still officially maintains a No First Use (NFU) policy, the country is very reluctant to respond to

bilateral initiatives and has stated they will only consider such talks only when the U.S. and Russia commit to deep cuts of their respective nuclear arsenals. (Kimball, 2021)

Meanwhile, there is no internationally agreed definition of lethal autonomous weapon systems. The U.S. has publicly committed to maintain the responsibility of human for the use of AI capabilities; however, the policy language remains vague, as it only stated that DoD personnel need to “exercise appropriate levels of judgment” with regard to these capabilities. (Department of Defense, 2023)

Internationally speaking, the UN Convention on Certain Conventional Weapons (UNCCW) incorporated a formal “Group of Governmental Experts” (GGE) to examine the technological and ethical issues surrounding lethal autonomous weapons. (Congressional Research Service, 2022) Additionally, numerous nongovernmental organizations have voiced concerns of the so-called “killer robots,” with some calling for a preemptive ban on lethal autonomous weapons due to concerns about risk, accountability, security, and legality. (Amnesty International) In short, multilateral engagement on both nuclear arms control and military automation is not at all novel, but we have yet seen the marriage of these two issues at the highest level.

4.4 Verification

Verification refers to the collection and analysis of data to determine if a state (or other party) is in compliance with a set of predefined measures. The United Nations Institute for Disarmament Research describes three primary objectives of verification: to detect non-compliance, deter non-complier (through threat of detection), and build confidence among signatories of an agreement (Dahlman, 2010). Traditional verification measures involve observables such as physical structures (e.g., missiles) or measurable qualities (e.g., radiation measurements). However, there is no nuclear arms control-related verification for software. In 2015, the Underhanded C hacking competition and the Nuclear Threat Initiative collaborated on a contest that demonstrated the ease with which hidden switches could be hidden in arms control verification software programs (Stoutland, 2015). Verification can occur in multiple ways.

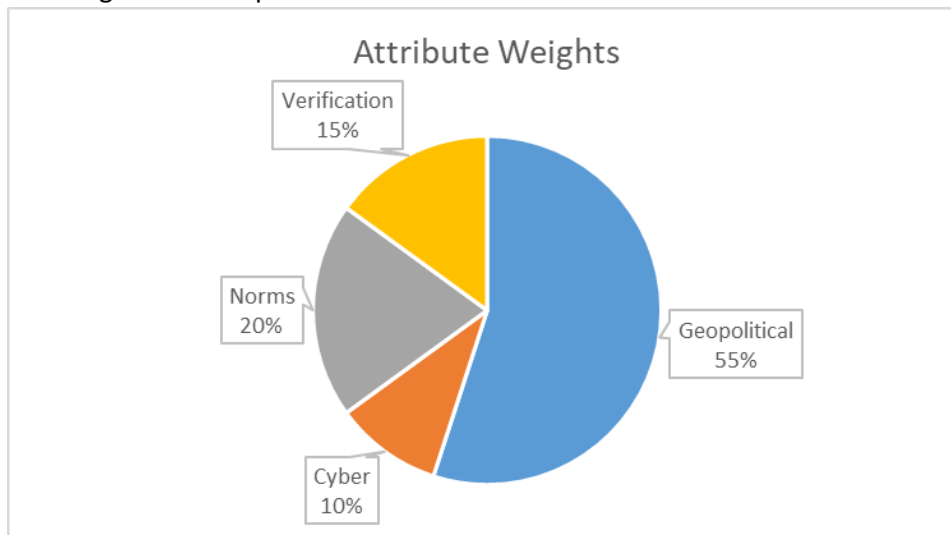
First, verification can occur through **unilateral measures** called “National Technical Means” (NTM) through which a state uses its resources to confirm specific aspects of an agreement. Examples of this type of unilateral verification are embodied in the Open Skies Treaty, in which each party accepts a specified number of overflights per year by observation aircraft (Treaty on Open Skies, 1992). Verification can also take place directly through the members of an agreement, such as **bilateral inspections** between the US and Russia in arms control agreements such the currently paused New START agreement (Gottemoeller, 2020) or between Argentina and Brazil in the Brazilian-Argentine Agency for Accounting and Control of Nuclear Materials (ABACC) (SBSCC, Goldembery, Alvim, & Mafra, 2018). Verification can also take place directly through the members of an agreement, such as **bilateral inspections** between the US and Russia in arms control agreements such the currently paused New START agreement (Gottemoeller, 2020) or between Argentina and Brazil in the Brazilian-Argentine Agency for

Accounting and Control of Nuclear Materials (ABACC) (SBSCC, Goldemberry, Alvim, & Mafra, 2018).

Third-party verification regimes are also common, such as the International Atomic Energy Agency's verification mission under the Nonproliferation Treaty (United Nations Office for Disarmament Affairs, n.d.). The Comprehensive Test Ban Treaty (CTBT) uses a hybrid between a third party verification body (Comprehensive Test Ban Treaty Organization) that monitors the data from the International Monitoring System, but also relies on a network of inspectors designated from member states (Preparatory Commission for the Comprehensive Test Ban Treaty Organization, 1996).

Some agreements, on the other hand, have **no formal verification** mechanisms. They might consist of unilateral declarations such as the Presidential Nuclear Initiatives, in which the United States and Soviet Union each announced unilateral nuclear arms reductions but were not accompanied by verification regimes (Corin, 2004). Traditional verification measures involve observables such as physical structures (e.g., missiles) or measurable qualities (e.g., radiation measurements). However, there is no nuclear arms control-related verification for software. In 2015, the Underhanded C hacking competition and the Nuclear Threat Initiative collaborated on a contest that demonstrated the ease with which hidden switches could be hidden in arms control verification software programs (Stoutland, 2015).

We weighted the importance of the attributes as follows:



To score the policy options on each attribute, we used a Likert scale from one to five, where 1 represents the hardest or riskiest option, and 5 represents the least risky or easiest option. We used consensus-based decision making for the weighting of the attributes and scoring of the policy options against those attributes. We discussed the weights and scores as a team until we all agreed on a weight or score and used that as our result. While we could have individually scored the regime and calculated the mean of the weights or scores, we thought that consensus-based scoring was less likely to inadvertently mask a disagreement among team members.

5.0 Analysis

A table describing the discussion behind our consensus-based scoring for each attribute is provided below. The score assigned to each attribute is listed in () below the attribute name, and the discussion notes are provided in the cells.

5.1 Status Quo

Geopolitical difficulty (3)	Cyber security (4)	Norms (4)	Verification (5)
Technology is developing faster than we are talking to international parties. This could lead to an autonomy arms race. Maintaining the status quo would be easy but would result in higher geopolitical risk.	The risks related to verification are null because this option does not require limits, but if autonomy development goes unchecked, it could result in higher risk due to our limited understanding on how to secure this technology.	Current policies do not constrain autonomy integration or development. Going in the direction of full autonomy for lethal systems would break the current policy of direct Presidential authorization for nuclear weapons launch.	Because there is no agreement or statement there is no expectation of verification activities.

5.2 Unilateral Statement

Geopolitical difficulty (3)	Cyber security (5)	Norms (3)	Verification (5)
Unilateral limits could make the United States vulnerable, since it would reduce only our domestic development. It would be low difficulty, but there could be geopolitical risk from falling behind in an autonomy arms race.	Unilateral limits could reduce internal cyber risk by limiting complexity of the nuclear command and control system. Unilateral limits would not introduce verification, which would maintain a low cyber risk.	Consistent with DOD's clarification on rules of use on autonomous lethal weapons in battle that only humans will make final decision on deploying lethal action. There is precedent for the United States making unilateral limits to nuclear capabilities.	Unilateral statements are not associated with verification measures historically.

5.3 Bilateral Agreement with Russia

Geopolitical difficulty (2)	Cyber security (4)	Norms (4)	Verification (3)
Current relations with Russia are strained due to the war in Ukraine, and the suspension of New START reduces communication channels.	Sharing information about our nuclear command and control systems with Russia could introduce a calculated risk.	Bilateral treaties with Russia are the norm for reducing risk from nuclear weapons. Including autonomy technologies would represent new scope.	Verification of computer codes would be a new area, but there might be other aspects to verification related to infrastructure, operations, etc. that could be developed to

			support bilateral verification measures.
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5.4 Bilateral Agreement with China

Geopolitical difficulty (2)	Cyber security (3)	Norms (1)	Verification (3)
Given China's development, a bilateral agreement would be attractive to the United States, so we do not get left behind. This will be more difficult to implement, given China's historical reticence to participate in arms control based on their smaller nuclear arsenal.	Sharing information about our nuclear command and control systems with China could be a calculated risk, but with more uncertainty given less experience sharing sensitive nuclear weapons information with China.	There is no precedent for bilateral engagement with China on nuclear arms control.	Verification of computer codes would be a new area, but there might be other aspects to verification related to infrastructure, operations, etc. that could be developed to support bilateral verification measures.

5.5 Multilateral, Non-binding Agreement

Geopolitical difficulty (3)	Cyber security (5)	Norms (5)	Verification (5)
Might be difficult to have China and Russia agree but they may be open to a conversation in the context of a non-binding agreement.	Likely minimal information sharing, combined with limitation of the three top players' integration of autonomy into nuclear weapons systems.	The UN convention on conventional weapons has formal discussions of government experts to examine technical aspects that could serve as a historical precedent and framework.	Since an agreement would not be legally binding, we might not require verification measures. Countries could write letters each year to discuss their efforts as a transparency method like current UNSCR 1540 activities.

5.6 International Binding Treaty

Geopolitical difficulty (1)	Cyber security (3)	Norms (3)	Verification (1)
This would be incredibly difficult given the current geopolitical tensions. While other countries might sign on, Russia, China, and other nuclear armed nations are unlikely to participate.	Most difficult. This would involve the largest number of parties having access to or visibility into nuclear launch codes, which is risky. However, if a treaty could successfully limit autonomous	There is significant precedent for international treaties on nuclear nonproliferation, but none have addressed automation, especially in the military context. The	Much harder in an international binding treaty, because getting a competent body who is able to make verification decisions will be more difficult and there is a risk

	nuclear launch systems around the world, that could contribute to reduced cyber risks overall.	United States is promoting global governance framework with 12 point non legal binding norms, that could be a good starting point.	of exposing not just the code (cyber risk) but also potentially nuclear weapons related information.
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Cost was analyzed but due to the limited information of Russia and China budgets we were only able to analyze cost within the United States context.

6.0 Findings

The final scores of each of the policy options is presented below in Figure 1. Based on the scoring above, we found that the multilateral, non-binding treaty had the most utility. The status quo and unilateral limitations were scored just below the multilateral agreement, driven by the significant geopolitical hurdles involved in getting the United States, Russia, and China to negotiate together. While one might ask why a multilateral agreement would be worth that effort given the relatively close score of the status quo, we think that the global risks resulting from integration of autonomous nuclear weapons launch capabilities in the future is too high to do nothing. Bilateral treaties with Russia and China both scored relatively low due to geopolitical tensions and the challenges associated with verifying the status/role of autonomy in a nuclear weapons system, and cyber threats introduced through verification measures. The international, binding treaty scored the lowest. While this is the “gold standard” of nuclear nonproliferation and arms control agreements, binding limits with invasive verification measures seemed untenable given the current geopolitical climate. And if such a treaty were negotiated without Russia, China, or other nuclear-armed states, it would not achieve the intended purpose of limiting autonomous nuclear weapons launch.

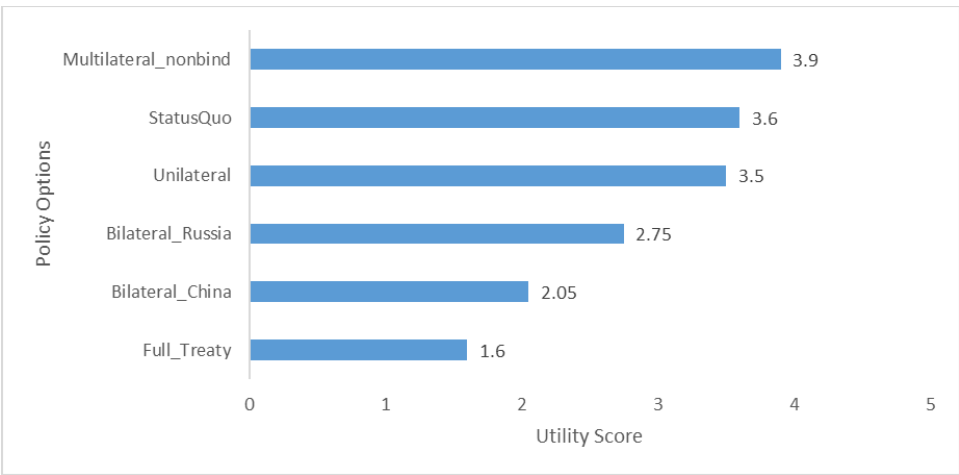


Figure 1. Nonproliferation Regime Utility Scores.

7.0 Implications, Conclusions, and Future Research

To conclude, the non-binding multilateral engagement is the most feasible approach to kickstart conversations among major nuclear states on the subject of limiting autonomous nuclear weapons launch and for the international community to voice concerns. At the same time, discussions on the broader implications of incorporating AI in military use could potentially be complementary to cultivate international norms that benefit nuclear arms control. A good example is the Responsible AI in the Military Domain Summit held in Hague in early 2023. The Summit brought together different stakeholders including governments, militaries, researchers, the private sector, and civil society to consider opportunities and risks associated with military AI, as well as the potential for regulation in this area. (Nadibaidze, 2023) Following the Summit, the U.S. introduced the Political Declaration on Responsible Military Use of Artificial Intelligence and Autonomy, in which states are expected to “maintain human control and involvement for all actions” critical to “executing decisions concerning nuclear weapons employment.” (Department of State, 2023) Granted, this outline does not make AI safer with the stroke of a pen, as endorsing states were only to adhere to “best practices” in policymaking. Nevertheless, it does represent a broadening of U.S. policies and continuation of international conversations on AI.

On the other hand, the rapid technological change and shift in geopolitical landscape are making concerns surrounding lethal autonomous weapons and military AI more pressing, especially because more states are either actively incorporating the latest generation of AI into military use or looking to develop AI capability. At the same time, it also creates both opportunities and challenges for researchers to examine the international security implications of greater use of AI (including generative AI), especially for international relations theory and situations such as deterrence and escalation during conflict. For global governance, the emerging dynamics of military AI could potentially generate new clusters of research on issues of explainable AI and enhancing trust of machine-dominated systems to delineate techniques and policies that reduce risks associated with military AI. Meanwhile, the cybersecurity risks associated with AI could also make legal-binding governance framework more challenging, as countries would have to be more transparent about source data and code during the verification process. Therefore, more innovative ways to enforce binding treaties are also critical for future regimes.

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